

FORM TP 2012147



TEST CODE **02207020**

MAY/JUNE 2012

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

ADVANCED PROFICIENCY EXAMINATION

BIOLOGY

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) 12.02.02
Figure 1 is an incomplete diagram summarising the main steps in the Calvin cycle.

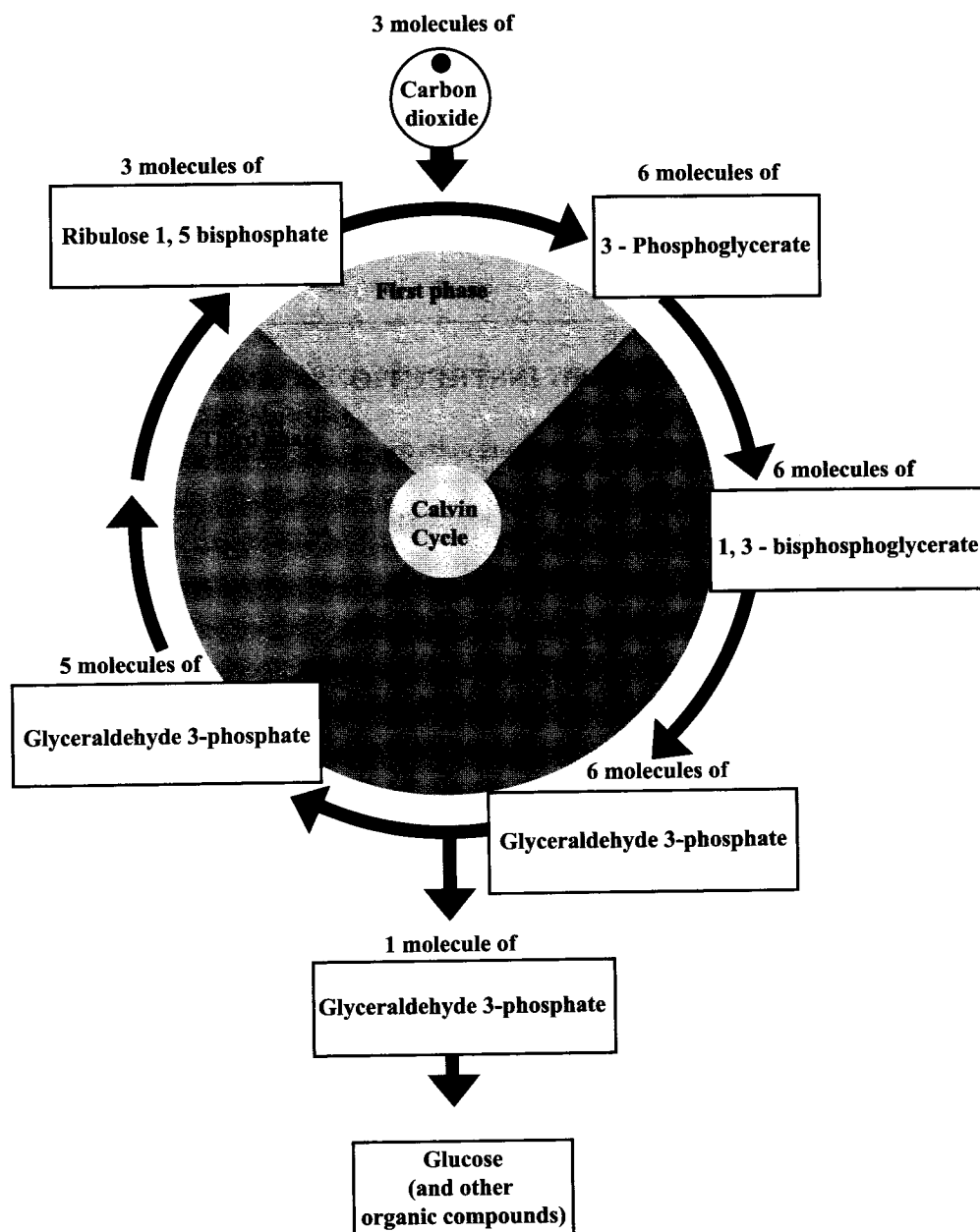


Figure 1. The Calvin cycle

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- (i) On Figure 1, use a labelled arrow to indicate EACH of the following:
- a) The **stages** at which ATP is used and ADP released, including the number of molecules involved
 - b) The **stage** at which NADPH is used and NADP⁺ released, including the number of molecules involved **[3 marks]**
- (ii) The series of reactions in the Calvin cycle may be grouped into three main phases as shown by the pie sections in Figure 1. Outline the MAIN purpose of EACH phase.

First phase: _____

Second phase: _____

Third phase: _____

[3 marks]

- (iii) The Calvin cycle and Krebs cycle are both metabolic pathways.

State ONE difference between these two metabolic pathways. **[NOTE: Do NOT state site of pathways as a difference.]**

[2 marks]

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- (b) Figure 2 shows the results of an experiment to investigate whether the use of pyruvate supplements enhances the aerobic performance of well-trained male cyclists. Seven well-trained male cyclists were given either placebo capsules (gelatin and glucose) or calcium pyruvate capsules for a period of seven days. Subjects cycled for a period of time until exhaustion. Whole-blood lactate and glucose concentrations were measured at 15-minute intervals during the period of aerobic exercise.

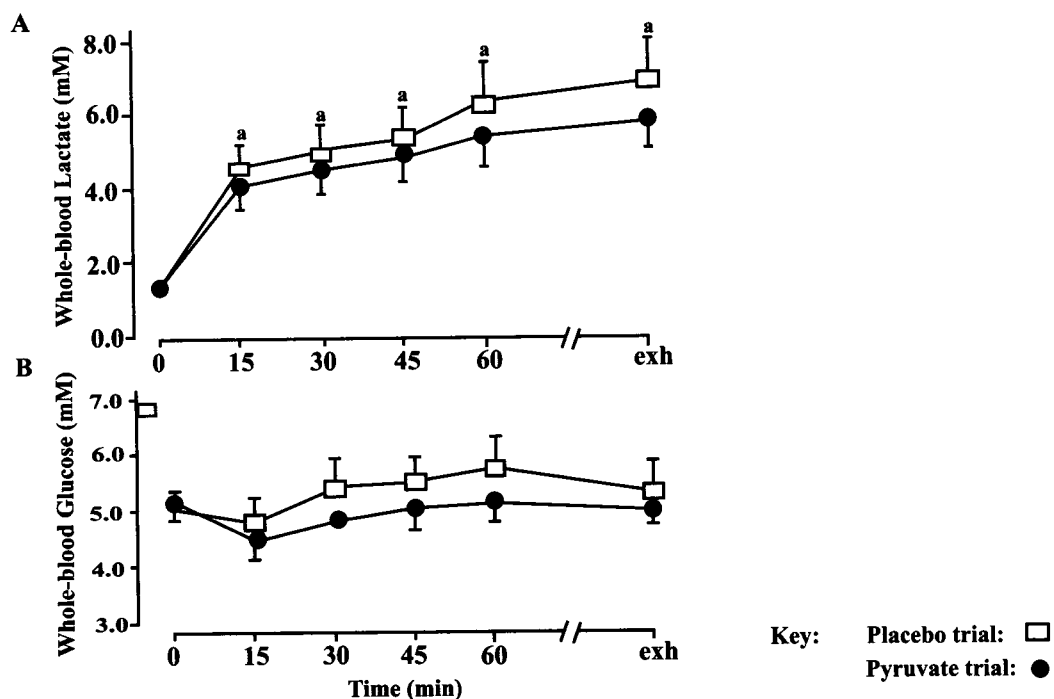


Figure 2. Whole-blood lactate (A) and glucose (B) concentrations during aerobic cycling after 7 days of consumption of placebo or calcium pyruvate.

Taken from: <http://jap.physiology.org/cgi/content-nw/full/89/2/549/T1>

- (i) Describe the changes in blood lactate and glucose levels for cyclists who used pyruvate supplements over the exercise period.

Lactate: _____

Glucose: _____

[4 marks]

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- (ii) Account for the trend observed with respect to the levels of lactate in the blood during the first 60 minutes of the cycling activity.

[2 marks]

- (iii) Based on the data shown in Figure 2, suggest what conclusion may be drawn with respect to the effect of pyruvate supplements on aerobic performance.

[1 mark]

Total 15 marks

2. (a) 2012.02.02
Figure 3 is a diagram of a longitudinal section through a mammalian heart, showing the main chambers and the electrical system.

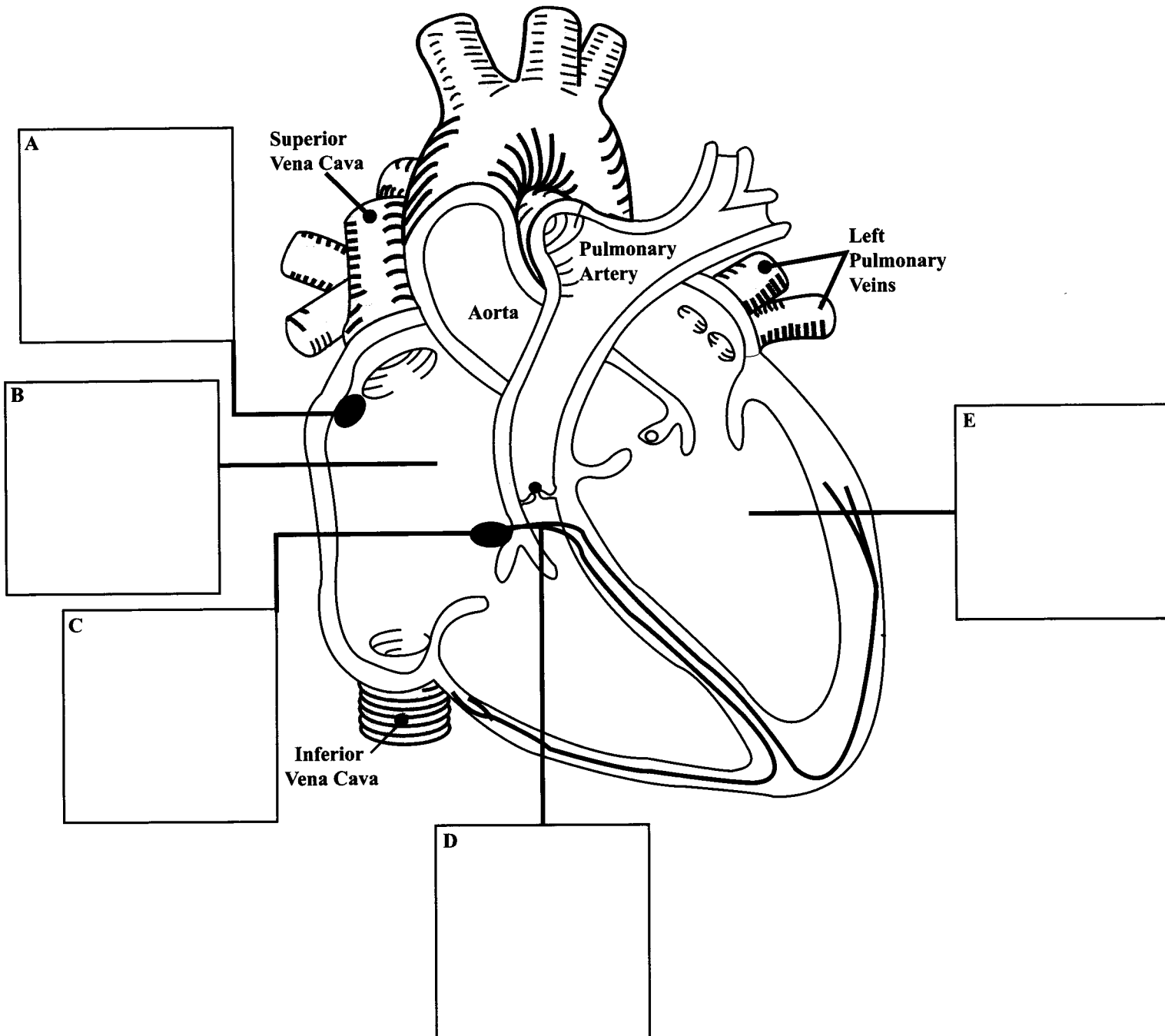


Figure 3. Longitudinal section through a mammalian heart

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- (i) In the boxes in Figure 3, use annotated labels to describe the structure of A, B, C, D and E. **[5 marks]**

- (ii) State the role of EACH of the following in the functioning of the heart:

A _____

C _____

D _____

[3 marks]

- (iii) State the role of the autonomic nervous system in the functioning of the heart.

[1 mark]

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- (b) Figure 4 is a schematic representation of the transverse section through a dicotyledonous root.

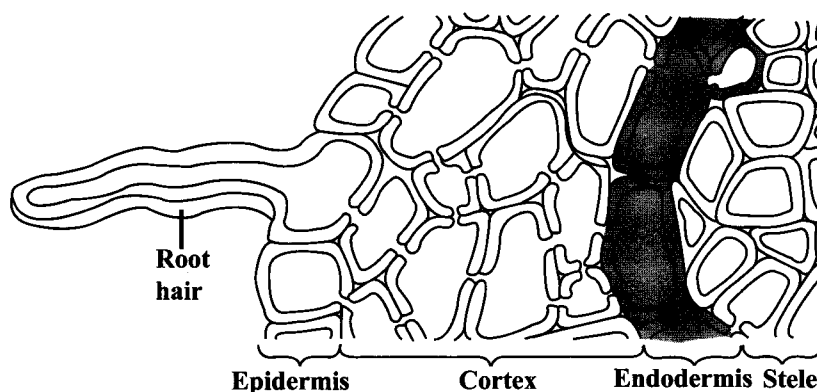


Figure 4. Schematic representation of the transverse section through a dicotyledonous root

- (i) Use arrows labelled A and B to indicate on Figure 4, TWO separate locations where the movement of ions across the root is by active transport. **[2 marks]**
- (ii) For EACH of the two locations identified in (b) (i) above, outline the movement of ions into and out of the cells.

A _____

B _____

[2 marks]

- (iii) Explain the relationship between the movement of ions and the movement of water across the root.

[2 marks]

Total 15 marks

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NOTHING HAS BEEN OMITTED.

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3. (a) ^{12.02.02} Table 1 is a summary of the leading causes of mortality in CAREC member countries for selected years over the period 1985 to 2000.

**TABLE 1: LEADING CAUSES OF MORTALITY (RATES PER 100 000) BY YEAR
FOR ALL CAREC MEMBER COUNTRIES**

Cause of Death	1985	1990	1995	2000
Heart disease	107.2	109.5	114.0	102.5
Cancers	81.1	79.3	88.5	64.9
Diabetes	36.2	50.0	51.9	64.6
Hypertension	30.6	30.9	31.4	39.8
Accidents	25.4	22.4	26.1	39.3

- (i) On the grid provided on page 11, plot the data for **Accidents** and **Hypertension** for the years 1985, 1990, 1995 and 2000 in the form of a bar graph. **[5 marks]**
- (ii) Comment on the trends in mortality rates recorded for **Cancers** and **Diabetes** over the 16-year period.

Cancers: _____

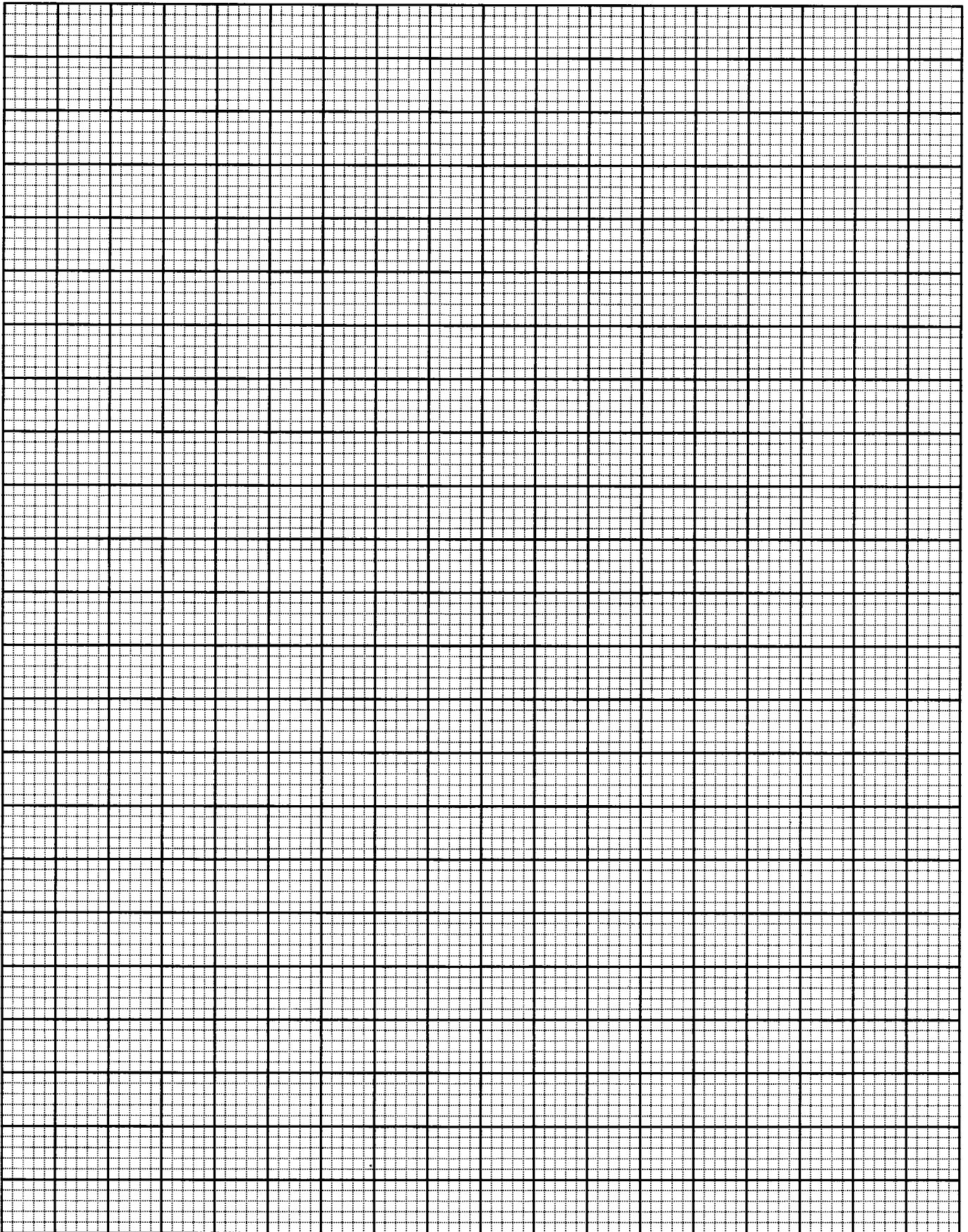
Diabetes: _____

[2 marks]

- (iii) Based on the data shown in Table 1, deduce the leading cause of death in CAREC member countries.

[1 mark]

Grid for Question 3 (a) (i)



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- (b) Figure 5 is a diagrammatic representation of the daily recommended food portions for a healthy balanced diet.

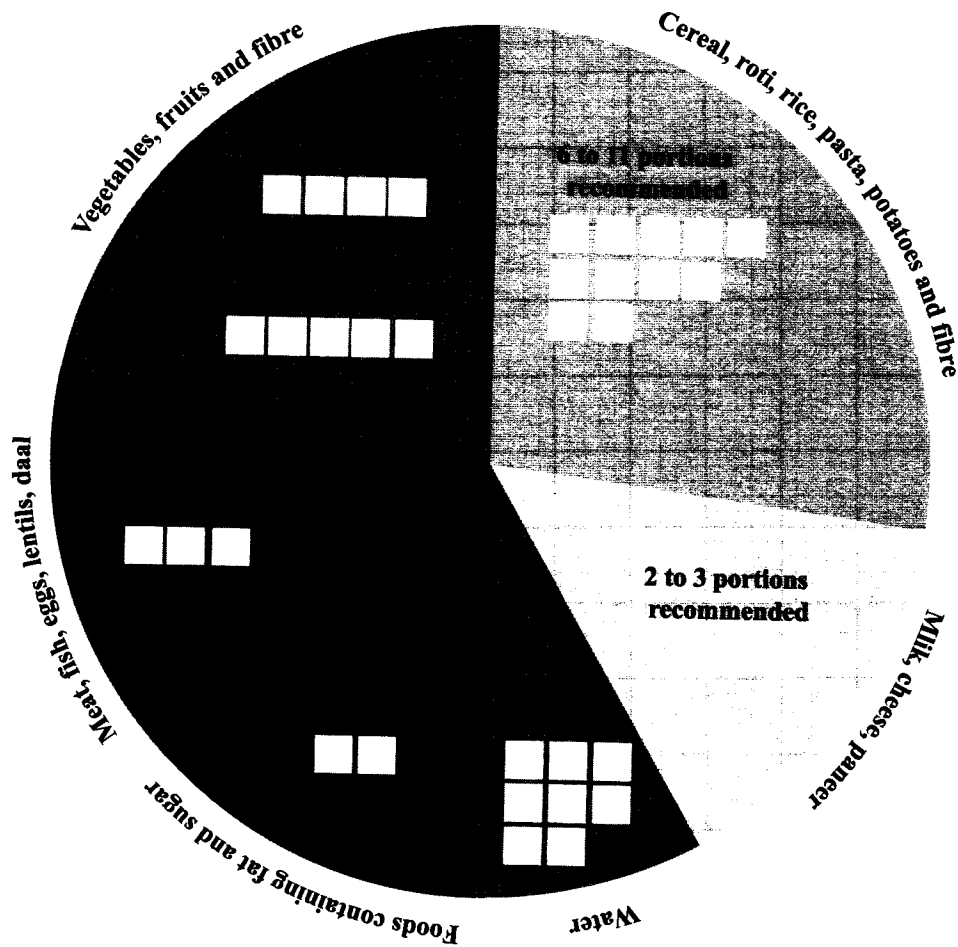


Figure 5. Daily recommended food portions for a healthy balanced diet

- (i) Explain what is meant by the term 'balanced diet'.

[2 marks]

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(ii) With reference to the information presented in Figure 5, state

a) TWO MAJOR categories of food in a balanced diet

[2 marks]

b) ONE food group which should be consumed in moderation.

[1 mark]

(iii) Comment on ONE possible health consequence of regularly consuming fats in excess of the daily requirement.

[2 marks]

Total 15 marks

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SECTION B

Answer ALL questions in this section.

Write your answers in the spaces provided at the end of each question.

4. (a) (i) Explain what is meant by the terms 'species diversity' and 'stability' of a community. [4 marks]
- (ii) Explain why species diversity is positively correlated with ecosystem stability. Your answer should make reference to resistance and resilience of the communities in the ecosystem. [5 marks]
- (b) (i) Define the term 'food chain' in terms of energy flow. [2 marks]
- (ii) Explain why a food web better describes energy flow in an ecosystem than does a food chain. [4 marks]

Total 15 marks

Write the answer to Question 4 here.

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Write the answer to Question 4 here.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper appears to be a standard notebook page, possibly from a composition book. The edges of the paper are slightly irregular, suggesting it might be a scan of a physical document. There is no handwriting or other markings on the page.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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- Total 15 marks**

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Write the answer to Question 5 here.

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6. (a) (i) Explain the term 'complement', and outline its role in the body's response to invasion by foreign organisms. **[6 marks]**
- (ii) Entry of foreign organisms into the body, for example, when there is a break in the skin, is often followed by redness, warmth and swelling at the site of entry.
- Discuss why EACH of these responses is an indication that the body is attempting to eliminate the foreign organisms. **[3 marks]**
- (b) Discuss THREE ways in which smoking cigarettes consistently over a prolonged period may result in breathlessness or difficulty in breathing. Refer to emphysema and bronchitis in your answer. **[6 marks]**

Total 15 marks

Write the answer to Question 6 here.

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IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.